# Christina Pizzonia

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Results-driven EE student with a focus in signal processing and digital design. Interested in software development and digital logic design/verification. Over 4 years of experience in research and programming.

## EDUCATION

**BASc, Electrical Engineering + PEY Co-Op**, University of Toronto 2022 - 2027

- Relevant coursework: Computer organization, digital systems, software design
- Dean's Honour List (cGPA: 3.96/4.00)

## SKILLS

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|---|---|
| **Languages** | Verilog, C/C++, Python, MATLAB |
| **Tools** | Quartus/Vivado, ModelSim/Vitis Unified IDE, Altium, Git, AutoCAD, Microsoft Office |
| **Soft Skills** | Written and oral communication, project management, technical research |

## EXPERIENCE

**Deep Learning Research Intern** May 2024 - Sept 2024
Genov Lab *Toronto, ON*

- Received an NSERC-USRA grant to develop a hardware-efficient, spiking neural network in Python for epileptic seizure prediction, to be implemented on an FPGA
- Performed a literature review on current deep learning algorithms for seizure prediction, and proposed a quantitative method to assess the implementability of algorithms in hardware
- Created custom IPs, modified pre-existing test-bench suites (Verilog) and worked with oscilliscopes to test incoming ASICs for functionality

**Summer Research Student** May 2022 - Aug 2022
Passeport Lab *Toronto, ON*

- Processed stormwater samples via density separation and organic digestion (in accordance with lab SOPs) and identified microplastics via FTIR spectroscopy to provide data for publication
- Analyzed microplastic data in Microsoft Excel to determine sources of error & improve accuracy of counting methods by $\approx 50\%$
- Performed a literature review on contaminant extraction and presented results to 2 academic labs

**Tutor** Mar 2021 - Present
Self-Employed *Toronto, ON*

- Have taught over 50 high-school and university students in calculus and improved scores by $> 25\%$
- Have grown a regular client-base from 0 to 10 students through targeted advertising and positive referrals

## PROJECTS

**Discrete FT Visualizer [GitHub]:** Programmed a Nios II processor on a DE1-SoC FPGA in C to compute and display the discrete Fourier Transform from either input audio or pre-selected waveforms on a monitor.

**HerWay [GitHub]:** Created an interactive map and GUI focused on women's safety in C++. Multi-target Dijkstra's, a greedy algorithm and 2-opt with multi-threading was implemented for finding optimal routes.

**Bindicator! [GitHub]:** Used waste collection data from OpenData to program a microcontroller to automatically update an LCD & LEDs with the waste type(s) being collected based on current time and location.

**Autonomous Robot [GitHub]:** Designed and programmed an autonomous, line-following robot using Fusion360/AutoCAD and microcontrollers to navigate a track using a bang-bang control algorithm